25900755 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE

EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBBT102 - BIOTECHNOLOGY AND HUMAN WELFARE

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer the Questions in One Word or Sentence

Answer **All** Questions

Each Question carries **1** mark

1. What are Biopesticides? [K] / [CO1]
2. What is a virus? [U] / [CO2]
3. How can GM crops affect biodiversity? [U] / [CO2]
4. What is Anthrax? [K] / [CO1]
5. What is a compost? [U] / [CO2]
6. Define genetically modified crops [U] / [CO2]
7. What is gene-therapy? [K] / [CO1]
8. Define crop management [U] / [CO2]
9. What are bioreactor applications in biotechnology? [A] / [CO3]
10. What are Patents? [K] / [CO1]

[10x1 = 10]

Part B

Short Answer Questions

Answer any **4** Questions

Each Question carries **3** marks

11. Why is timing important in harvesting? [U] / [CO2]

12. Which are the major products of diary biotechnology? [An] / [CO3]

13. How does white biotechnology impact sustainability? [K] / [CO1]

14. What is biocontrol of weeds? [U] / [CO2]

15. What are some applications of biotechnology in industry? [An] / [CO3]

16. Write down the major periods in history of biotechnology [K] / [CO1]

[4x3 = 12]

Part C

Short Essay Questions

Answer any **3** Questions

Each Question carries **6** marks

17. Elaborate on the methods of antibiotic production [A] / [CO3]

18. Write down the management of E-wastes [E] / [CO4]

19. How does Gold Biotechnology contribute to modern biological research and medicine? [K] / [CO1]

20. What is activated sludge, and how does it work in wastewater treatment? [E] / [CO4]

21. What are the challenges in addressing the health-environment nexus, and how can these challenges be overcome? [U] / [CO2]

[3x6 = 18]

Part D

Essay Questions

Answer any **1** Question

Each Question carries **10** marks

22. How has modern biotechnology revolutionized agriculture compared to conventional methods? [K] / [CO1]

23. Explain the principle and application of biofilters in wastewater treatment. [E] / [CO4]

[1x10 = 10]